Supplementary Material for "Cast-Break-Free Demand Response of Electric Arc Furnace Fleets: Exact Counting Polytopes and Certified Rolling Dispatch"

Kaidi Huang
Department of Electrical Engineering, Tsinghua University, Beijing, China

This document proves Theorems 1 to 4, with the constants and the hard instances of Theorem 3. It also gives the details that the paper states in words: the boundary rows of (3), the arc LP and the LP of Algorithm 1, and the example cited after Theorem 1. Equations without the prefix S are numbered as in the paper.

I. Boundary Rows of the Counting Polytope

Each last start of the live state is placed so that its heat or cast ends at the actual tap or cast end. Heats are indexed by start from these last starts, so heats $1,\dots,n$ are the last heats of the furnaces idle at the notice and casts $1,\dots,N^{\mathrm{C}}$ are the last casts of the casters. Under first-in first-out casting, the Q waiting ladles feed casts $N^{\mathrm{C}}+1,\dots,N^{\mathrm{C}}+Q$, and the ladle of heat $k>n$ feeds cast $k+q$, with $q=Q+N^{\mathrm{C}}-n$. The counters before $t=0$ are fixed, (3d) is imposed for $t\ge 0$, (3e) for $x(t)>n$, and

$$y(\tau_j+w)\ge N^{\mathrm{C}}+j,\qquad j=1,\dots,Q, \tag{S1}$$

where $\tau_j\le 0$ is the tap of the j-th oldest waiting ladle. Beyond H, (3b) and (3c) use $x(\min\{\cdot,H\})$ and $y(\min\{\cdot,H\})$, and the other rows are dropped.

II. Proof of Theorem 1

For integer counters, let $s_k=\min\{t: x(t)\ge k\}$, $z_k=\min\{t: y(t)\ge k\}$ and $v_k=s_k+d(s_k)$. Then $x(t)\ge k \Leftrightarrow s_k\le t$, $y(t)\ge k\Leftrightarrow z_k\le t$, and, as $t+d(t)$ is non-decreasing, $x(\theta(t))\ge k\Leftrightarrow v_k\le t$.

Lemma S1. *For integer counters satisfying* (3a)*, rows* (3b) *to* (3f) *hold if and only if, for all starts in the horizon,*

$$s_{k+N^{\mathrm{E}}}\ge v_k+g, \tag{S2a}$$
$$z_k+c(z_k)\le z_{k+N^{\mathrm{C}}}\le z_k+c(z_k)+b, \tag{S2b}$$
$$v_k\le z_{k+q}\le v_k+w,\quad k>n. \tag{S2c}$$

Proof. Rows (3b), (3c), (3f), (3d) and (3e) at $t=s_k$, z_k, z_k, z_{k+q} and s_k give

$$x(v_k+g-1)\le k-1+N^{\mathrm{E}},\ y\big(z_k+c(z_k)-1\big)\le k-1+N^{\mathrm{C}},$$
$$y\big(z_k+c(z_k)+b\big)\ge k+N^{\mathrm{C}},\quad x\big(\theta(z_{k+q})\big)\ge k,$$
$$y(v_k+w)\ge k+q,$$

which are (S2a) to (S2c). Conversely, for (3b) at t, let $k=x(t-1)+1$. Then $s_k\ge t$, and (S2a) gives $s_{k+N^{\mathrm{E}}}\ge t+d(t)+g$, that is, $x(t+d(t)+g-1)\le x(t-1)+N^{\mathrm{E}}$. Rows (3c), (3f), (3d) and (3e) follow in the same way with $k=y(t-1)+1$, $y(t)$, $y(t)-q$ and $x(t)$, and (3d) holds trivially if $y(t)\le N^{\mathrm{C}}+Q$, since $x(\theta(t))\ge n$ for $t\ge 0$. This proves the lemma. □

Proof of Theorem 1. Let a schedule be zero-break. A furnace that starts a heat at $s\in[t,\,t+d(t)+g-1]$ starts its next heat at $s'\ge s+d(s)+g\ge t+d(t)+g$ by (1d), so each furnace starts at most one heat in this window, which gives (3b), and (1e) gives (3c) and (3f) in the same way. The $y(t)-N^{\mathrm{C}}$ casts started by t after the last casts need distinct ladles, waiting or tapped in $[0,t]$, and by (1f) the waiting ladles and those of the heats $k>n$ started by t are cast by $t+d(t)+w$. Hence

$$y(t)-N^{\mathrm{C}}\le Q+x\big(\theta(t)\big)-n,$$
$$y\big(t+d(t)+w\big)-N^{\mathrm{C}}\ge Q+x(t)-n,$$

which are (3d) and (3e), and (S1) holds likewise. As the counters do not decrease after H, the capped rows also hold. Conversely, the starts of an integer point of $\mathcal{F}$ satisfy (S2) by Lemma S1. Under the round-robin assignment, heat $k+N^{\mathrm{E}}$ is the next heat on the furnace of heat k, cast $k+N^{\mathrm{C}}$ is the next cast on the caster of cast k, and cast $k+q$ receives the ladle of heat k. Then (S2) and (S1) are (1d) to (1f), so the round-robin schedule is zero-break. Hence the integer points of $\mathcal{F}$ are exactly the counters of the zero-break schedules, each realized by the round-robin assignment.

Each row of $\mathcal{F}$ is a difference constraint with an integer right side, so the constraint matrix of $\mathcal{F}$ is totally unimodular by [1]. As $\mathcal{F}$ is bounded by (3b) and (3c), it is an integral polytope. This proves Theorem 1. □

As (1c) is linear, it maps $\mathcal{F}=\mathrm{conv}(\mathcal{F}\cap\mathbb{Z})$ onto the convex hull of the image of $\mathcal{F}\cap\mathbb{Z}$, which by Theorem 1 is the set of block loads of the zero-break schedules. This proves the statement on block loads that follows Theorem 1 in the paper.

Example S1. This hull need not be a g-polymatroid. Let $N^{\mathrm{E}}=N^{\mathrm{C}}=1$, $d=2$, $g=0$, $c=3$ and $b=w=1$, with an idle furnace, no waiting ladle, a cast in progress that ends at $t=1$, $t_0=0$ and $H\ge 10$. Then $s_1=0$, $z_1=2$, $z_{k+1}-z_k\in\{3,4\}$,

$z_k - v_k \in \{0,1\}$ and $s_{k+1} \geq v_k$. Enumerating these schedules gives, in units of $P/3$, the load vectors (ℓ_1, ℓ_2, ℓ_3) of the first three blocks

$$(2,2,1),\ (2,2,2),\ (2,2,3),\ (2,3,1),\ (2,3,2),$$
$$(3,1,2),\ (3,1,3),\ (3,2,1),\ (3,2,2).$$

The function $\ell_1+2\ell_2+\ell_3$ equals 7 at $(2,2,1)$, with $s = (0,3,8)$, and at $(3,1,2)$, with $s = (0,2,6,9)$, and is at least 8 elsewhere. These two vectors thus span an edge of the projected hull in direction $(1,-1,1)$, whereas every edge of a g-polymatroid and of its coordinate projections is parallel to some e_j or $e_j - e_k$ [2], so the hull is not a g-polymatroid.

III. Proof of Theorem 2

An arc (s,v) is a heat started at s and tapped at v, admissible if $(v-s)\underline{P} \leq E \leq (v-s)\overline{P}$. At slot prices π_t, its cheapest power profile costs

$$\kappa(s,v) = \min_p \Big\{ \sum_{t=s}^{v-1} \pi_t p_t :\ \underline{P} \leq p_t \leq \overline{P},\ \sum_{t=s}^{v-1} p_t = E \Big\}. \quad \text{(S3)}$$

The arc LP takes one variable $\xi(s,v) \geq 0$ per admissible arc, counts the starts by $x(t) = \sum_{s \leq t} \sum_v \xi(s,v)$ and the taps by $\hat{x}(t) = \sum_{v \leq t} \sum_s \xi(s,v)$, replaces rows (3b), (3d) and (3e) by $x(t) - \hat{x}(t-g) \leq N^{\mathrm{E}}$, $y(t) - \hat{x}(t) \leq q$ and $y(t+w) - \hat{x}(t) \geq q$, keeps the cast rows (3a), (3c) and (3f), and minimizes $\sum_{(s,v)} \kappa(s,v)\xi(s,v)$ plus any cost linear in y.

With $p_t = \underline{P} + (\overline{P} - \underline{P})\omega_t$, (S3) becomes

$$\kappa(s,v) = \underline{P} \sum_{t=s}^{v-1} \pi_t + (\overline{P} - \underline{P})\, \phi\big(\{s, \dots, v-1\}, \nu_{v-s}\big), \quad \text{(S4)}$$

where $\phi(I,\nu) = \min\{\sum_{t \in I} \pi_t \omega_t : \omega \in [0,1]^I,\ \sum_{t \in I} \omega_t = \nu\}$ is convex in ν on $[0, |I|]$, $\nu_d = (E - d\underline{P})/(\overline{P} - \underline{P})$, and $\nu_{d+1} = \nu_d - \delta$ with $\delta = \underline{P}/(\overline{P} - \underline{P})$. The first term of (S4) is modular in (s,v) and cancels below.

Let $(s, v+1)$ and $(s+1, v)$ be admissible, $I = \{s+1, \dots, v-1\}$, $\nu = \nu_{v-s}$, and let α and α' be the weights of s and v in an optimal solution of $\phi(I \cup \{s,v\}, \nu - \delta)$. Then

$$\begin{aligned}
&\phi(I \cup \{s\}, \nu) + \phi(I \cup \{v\}, \nu) \\
&\quad \leq \pi_s \alpha + \pi_v \alpha' + \phi(I, \nu - \alpha) + \phi(I, \nu - \alpha') \\
&\quad \leq \pi_s \alpha + \pi_v \alpha' + \phi(I, \nu - \delta - \alpha - \alpha') + \phi(I, \nu + \delta) \\
&\quad = \phi(I \cup \{s,v\}, \nu - \delta) + \phi(I, \nu + \delta),
\end{aligned}$$

where the second step is convexity, as $\nu - \delta - \alpha - \alpha' \leq \nu - \alpha, \nu - \alpha' \leq \nu + \delta$ have equal sums and lie in $[0, |I|]$ because $\alpha + \alpha' \leq \nu - \delta$ and $(s+1, v)$ is admissible. By (S4),

$$\kappa(s,v) + \kappa(s+1, v+1) \leq \kappa(s, v+1) + \kappa(s+1, v). \quad \text{(S5)}$$

Admissible durations form an interval, so summing (S5) over $s_1 \leq s < s_2$ and $v_1 \leq v < v_2$ gives the Monge inequality $\kappa(s_1,v_1) + \kappa(s_2,v_2) \leq \kappa(s_1,v_2) + \kappa(s_2,v_1)$ for all admissible nested arcs (s_1,v_2) and (s_2,v_1).

Heats in progress are fixed and only shift the integer right sides of the rows. Moving $\varepsilon = \min\{\xi(s_1,v_2), \xi(s_2,v_1)\}$ from a nested pair to (s_1,v_1) and (s_2,v_2) keeps x, $\hat{x}$ and every row, does not raise the cost, and lowers $\sum \xi(s,v)(v-s)^2$ by $2\varepsilon(s_2 - s_1)(v_2 - v_1)$. Some optimal ξ thus has no nested pair, and its support orders as $a_1, \dots, a_R$ with non-decreasing starts and taps. For $u \in [0,1)$ and every row $a - a' \leq h$ with $h \in \mathbb{Z}$,

$$\lfloor a + u \rfloor - \lfloor a' + u \rfloor \leq \lceil a - a' \rceil \leq h, \quad \text{(S6)}$$

so the shifted floors of x, $\hat{x}$ and y keep every row [3]. With $M_j = \sum_{i \leq j} \xi(a_i)$, the k-th rounded start and tap both lie on a_{j_k}, $j_k = \min\{j : M_j \geq k - u\}$, so the rounding is a zero-break schedule on arcs of the support. For u uniform on $[0,1)$, arc a_j carries $\xi(a_j)$ heats on average and $\mathbb{E}\lfloor y + u \rfloor = y$, so some u gives an integral schedule that costs at most the LP value, which is therefore optimal. This proves Theorem 2. □

IV. Proof of Theorem 3

We prove Theorem 3 in a precise form. In part (i), at most $m-1$ plants are fractional at an optimal vertex of the fleet LP, and every $\Delta > P_{\max}\sqrt{(m-1)\ln m/2}$ is a certificate. Part (ii) holds already for single-furnace plants with start-dependent heat and casting times and arbitrary baselines: $C^\star - C^{\mathrm{LP}}$ can reach $\Omega(\sqrt{m}\, P_{\max})$, and for some constant $\epsilon > 0$, it is NP-hard to guarantee a peak within $\epsilon\sqrt{m}\, P_{\max}$ of $C^\star$.

1) Part (i): By (S6), the shifted floors (4) of a point of $\mathcal{F}_i$ lie in $\mathcal{F}_i \cap \mathbb{Z}$ and are zero-break schedules by Theorem 1. For u uniform on $[0,1)$, $\mathbb{E}\lfloor a+u \rfloor = a$, and each running count rounds to $\lfloor r_i(t) \rfloor$ or $\lceil r_i(t) \rceil$, so $\mathbb{E}\,\ell_{i,\beta} = \ell^{\mathrm{LP}}_{i,\beta}$ and $\ell_{i,\beta}$ ranges over an interval of length P_i. Markov's inequality, the independence of the shifts and the union bound give

$$\Pr\Big\{ \max_\beta \Big(\sum_i \ell_{i,\beta} - \bar{L}_\beta \Big) \geq C^{\mathrm{LP}} + \Delta \Big\} \leq U(\Delta, \lambda).$$

Let U_j be U with the shifts of the first j plants of $\mathcal{S}$ fixed. As U_j averages U_{j+1} over the next shift, a minimizing shift gives $U_{j+1} \leq U_j$, and $U_{|\mathcal{S}|} = \sum_\beta \exp\big(\lambda(\sum_i \ell_{i,\beta} - \bar{L}_\beta - C^{\mathrm{LP}} - \Delta)\big) < 1$ puts every block below $C^{\mathrm{LP}} + \Delta$. The floors of plant i change only where u_i crosses one minus the fractional part of one of the $O(m)$ counters in its block loads, so every expectation in (5) has $O(m)$ terms, and the rounding is polynomial.

At an optimal vertex, let G_i be the minimal face of $\mathcal{F}_i$ containing (x_i, y_i). The vertex is the only point of $G = \prod_i G_i \times \mathbb{R}$ near it at which the tight rows of (1b) are equalities, so $1 + \sum_i \dim G_i = \dim G \leq m$. As $\mathcal{F}_i$ is integral, $\dim G_i \geq 1$ for $i \in \mathcal{S}$, and $|\mathcal{S}| \leq m - 1$ [4]. By Hoeffding's lemma, each expectation in (5) is at most $e^{\lambda^2 P_i^2/8}$, so with $C^{\mathrm{LP}}_\beta \leq C^{\mathrm{LP}}$ and $\lambda = 4\Delta/\sum_{i \in \mathcal{S}} P_i^2$,

$$U(\Delta,\lambda) \leq m\, e^{-2\Delta^2/\sum_{i \in \mathcal{S}} P_i^2} \leq m\, e^{-2\Delta^2/((m-1)P_{\max}^2)}, \quad \text{(S7)}$$

which is below one for $\Delta > P_{\max}\sqrt{(m-1)\ln m/2}$, so every such Δ is a certificate. This proves part (i).

2) Part (ii): Given sets $T_1, \dots, T_M \subseteq \{1, \dots, N\}$, give T_k the blocks A_k of slots $6k$ to $6k+2$ and B_k of slots $6k+3$ to $6k+5$, so $m = 2M$, and give element i a plant with $N^{\mathrm{E}} = N^{\mathrm{C}} = 1$, power $P = P_{\max}$, $g = w = 0$, $b = 1$ and $d(s) = 3,4,7,6,5,4$ for $s \equiv 0, \dots, 5 \pmod 6$. Then $s + d(s)$ is non-decreasing and taps fall on slots $\equiv 3$ or $5 \pmod 6$. A cast started at $6k+3$ or $6k+5$ with $i \in T_k$ lasts until the same offset in the next

set containing i, and a lead-in before the window lets plant i choose the offset of its first cast.

As $w = 0$ and $b = 1$, each cast starts at the end of the previous one, so plant i keeps one offset, $\chi_i = +1$ for 3 and $\chi_i = -1$ for 5. A tap at $6k + 3$ needs a heat covering A_k, and a tap at $6k + 5$ the heat in slots $6k + 1$ to $6k + 4$, so every schedule of plant i loads each block at least as much as the one that adds $P(5 + \chi_i)/6$ to A_k and $P(1 - \chi_i)/3$ to B_k for each k with $i \in T_k$. With $\bar{L}_{A_k} = 5P|T_k|/6 - C_0$, $\bar{L}_{B_k} = P|T_k|/3 - C_0$ and $\sigma_k = \sum_{i \in T_k} \chi_i$, the excesses are $C_0 + P\sigma_k/6$ and $C_0 - P\sigma_k/3$. As the block loads of $\mathcal{F}_i$ form the convex hull of those of its zero-break schedules, the fleet LP may take $\chi \in [-1, 1]^N$, and it has $C^{\mathrm{LP}} = C_0$, attained at $\chi = 0$, since $\max\{\sigma_k/6, -\sigma_k/3\} \geq 0$. Hence

$$C^\star - C^{\mathrm{LP}} = \min_{\chi \in \{\pm 1\}^N} \max_k \max \left\{ \frac{P}{6}\sigma_k, -\frac{P}{3}\sigma_k \right\} \in \left[\frac{P}{6}, \frac{P}{3} \right] \mathrm{disc}(T). \tag{S8}$$

For a Hadamard matrix H of order $N = 2^r$ with first row all ones and $T_k = \{i : H_{ki} = 1\}$, every $\chi \in \{\pm 1\}^N$ with $\Sigma = \sum_i \chi_i$ has $\sigma_1 = \Sigma$, $(H\chi)_k = 2\sigma_k - \Sigma$ and $\|H\chi\|^2 = N^2$, so $\max_k |\sigma_k| \geq \max\{|\Sigma|, (\sqrt{N} - |\Sigma|)/2\} \geq \sqrt{N}/3$. By (S8), $C^\star - C^{\mathrm{LP}} \geq P\sqrt{N}/18 = \Omega(\sqrt{m}\, P_{\max})$. For set systems with $\Theta(N)$ sets on N elements, telling $\mathrm{disc}(T) = 0$ from $\mathrm{disc}(T) \geq c\sqrt{N}$ is NP-hard for some constant $c > 0$ [5]. By (S8), $C^\star = C^{\mathrm{LP}}$ in the first case and $C^\star \geq C^{\mathrm{LP}} + cP\sqrt{N}/6$ in the second. As $m = \Theta(N)$, there is a constant $\epsilon > 0$ with $\epsilon\sqrt{m} < c\sqrt{N}/6$, and a polynomial algorithm with a peak guaranteed within $\epsilon\sqrt{m}\, P$ of $C^\star$ would tell the two cases apart. Hence no such algorithm exists unless P = NP. This proves part (ii). □

V. Proof of Theorem 4

Lemma S2. *Let s_k be the k-th heat start. For all heat times within the bounds that agree with the observations up to slot t, the k-th tap lies in $[\underline{\eta}_t(s_k), \bar{\eta}_t(s_k)]$.*

Proof. The heats $k' \leq k$ tap by $s_{k'} + \overline{d}_t(s_{k'}) \leq \bar{\eta}_t(s_k)$, and the heats $k' \geq k$ tap at or after $s_{k'} + \underline{d}_t(s_{k'}) \geq \underline{\eta}_t(s_k)$, so at least k taps occur by $\bar{\eta}_t(s_k)$ and at most $k - 1$ before $\underline{\eta}_t(s_k)$. Hence the k-th tap lies in $[\underline{\eta}_t(s_k), \bar{\eta}_t(s_k)]$. □

With the loads of the executed slots fixed, and with a heat running at the notice drawing power until its latest tap and a later heat for its nominal time, Algorithm 1 solves, over the m_t blocks $\mathcal{B}_t$ that remain, the LP

$$\min \quad C + \frac{1}{m_t} \sum_{\beta \in \mathcal{B}_t} C_\beta - \gamma \sum_{i \in \mathcal{I}} \sum_{s \geq t_0 + 3m} \big(x_i(s) + y_i(s)\big) \tag{S9a}$$

$$\text{s.t.} \quad \textstyle\sum_{i \in \mathcal{I}} \ell_{i,\beta} - \bar{L}_\beta \leq C_\beta \leq C, \quad \beta \in \mathcal{B}_t, \tag{S9b}$$

$$(x_i, y_i) \in \mathcal{R}_{i,t}, \quad i \in \mathcal{I}, \tag{S9c}$$

whose objective lowers the peak and the mean block load and, through its last term with a small weight γ, starts heats and casts as early as possible after the window.

We say that a dispatch respects $\mathcal{R}_{i,t}$ if, whenever $\mathcal{R}_{i,t}$ is feasible, it executes the starts of slot t of an integer point of $\mathcal{R}_{i,t}$ with round-robin units and first-in first-out ladles.

Lemma S3. *If $\mathcal{R}_{i,0}$ is feasible and a dispatch respects $\mathcal{R}_{i,t}$, then $\mathcal{R}_{i,t}$ is feasible at every slot t and the dispatch is robust zero-break for plant i.*

Proof. Let p be an integer point of $\mathcal{R}_t$, executed with round-robin units and first-in first-out ladles, and fix heat times within the bounds that agree with the observations up to t. With s_k and z_k the starts of p, rows (7a) at $s = s_k$, (7b) at $s = z_{k+q}$ and (7c) at $s = s_k$ give

$$s_{k+N^{\mathrm{E}}} \geq \bar{\eta}_t(s_k) + g, \qquad \bar{\eta}_t(s_k) \leq z_{k+q} \leq \underline{\eta}_t(s_k) + w.$$

As heat k taps by $s_k + \overline{d}_t(s_k) \leq \bar{\eta}_t(s_k)$ and, by Lemma S2, the k-th tap lies in $[\underline{\eta}_t(s_k), \bar{\eta}_t(s_k)]$, these give (1d) and (1f), while (S1) covers the waiting ladles and (S2b) gives (1e). New observations give $\bar{\eta}_{t+1} \leq \bar{\eta}_t$, $\underline{\eta}_{t+1} \geq \underline{\eta}_t$ and $\bar{\theta}_{t+1} \geq \bar{\theta}_t$, so, the counters being non-decreasing, every row of $\mathcal{R}_{t+1}$, including the fixed starts, holds at the point p_t executed at t. Hence every $\mathcal{R}_{i,t}$ is feasible, and by the argument above, the executed schedule, equal to p_T at the last slot T, is zero-break at the realized heat times, for every realization within the bounds. This proves the lemma. □

By (S6), every combination of shifted floors that Algorithm 1 may execute lies in $\mathcal{R}_{i,t} \cap \mathbb{Z}$, so the algorithm respects $\mathcal{R}_{i,t}$, and by Lemma S3, every $\mathcal{R}_{i,t}$ stays feasible and the algorithm is robust zero-break. At an optimal vertex of (S9), all left rows of (S9b) are tight, as C_β has a positive cost in (S9a), and if $\mu \leq m_t$ right rows are tight, the face count of part (i) of Theorem 3 with the $1 + m_t$ variables C and C_β gives $1 + m_t + \sum_i \dim G_i \leq m_t + \mu$. At most $m_t - 1$ plants are thus fractional, the rest of that proof holds for the m_t remaining blocks with the peak of the vertex for C^{LP}, and the search accepts only lower peaks. This proves Theorem 4. □

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